

FCC §15.247 Pre-Compliance Test Procedures

US915 LoRa — 902–928 MHz ISM Band

DTS • FHSS • Hybrid Test Coverage

Standard: 47 CFR §15.247 & ANSI C63.10-2020

Band: 902–928 MHz ISM (US915)

Technology: LoRa — DTS, FHSS, Hybrid

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FCC §15.247 Pre-Compliance Conducted Test Procedures

US915 LoRa — 902–928 MHz ISM Band

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Reference Documents

- 47 CFR §15.247 (current as of 2/26/2026)
 - ANSI C63.10-2020 with Corrigendum 1 (2023)
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Section 1: Power Measurement Strategy — Average vs. Peak

3.1 Regulatory Basis

FCC §15.247(b) specifies maximum peak conducted output power for all system types. However, §15.247(b)(3) provides an alternative for digitally modulated systems:

47 CFR §15.247(b)(3): "For systems using digital modulation in the 902-928 MHz, 2400-2483.5 MHz, and 5725-5850 MHz bands: 1 Watt. As an alternative to a peak power measurement, compliance with the one Watt limit can be based on a measurement of the maximum conducted output power. Maximum Conducted Output Power is defined as the total transmit power delivered to all antennas and antenna elements averaged across all symbols in the signaling alphabet when the transmitter is operating at its maximum power control level."

This alternative is **only available in paragraph (b)(3)** — it does not appear in (b)(1) or (b)(2). Paragraph (b)(2), which governs FHSS systems in 902–928 MHz, specifies only "maximum peak conducted output power" with no average alternative.

3.2 Strategy for This Test Program

SYSTEM TYPE	FCC PARAGRAPH	POWER METHOD	LIMIT	BAND-EDGE ATTEN.
DTS (LoRa digital modulation)	§15.247(b)(3)	Average (chosen)	+30 dBm (1 W)	30 dB per §15.247(d)
FHSS (frequency hopping component)	§15.247(b)(2)	Peak (required)	+30 dBm (1 W) for ≥50 channels	20 dB per §15.247(d)
Hybrid (DTS + FHSS combined)	§15.247(f)	See note below	See note below	See note below

Hybrid Power Strategy

Per §15.247(f), hybrid systems are tested by decomposing into their two components:

47 CFR §15.247(f): "For the purposes of this section, hybrid systems are those that employ a combination of both frequency hopping and digital modulation techniques. The frequency hopping operation of the hybrid system, with the direct sequence or digital modulation operation turned-off, shall have an average time of occupancy on any frequency not to exceed 0.4 seconds within a time period in seconds equal to the number of hopping frequencies employed multiplied by 0.4. The power spectral density conducted from the intentional radiator to the antenna due to the digital modulation operation of the hybrid system, with the frequency hopping operation turned off, shall not be greater than 8 dBm in any 3 kHz band during any time interval of continuous transmission."

- **FHSS component** (DTS off): Tested for hopping parameters only (dwell time, channel count). No separate FHSS power test in §15.247(f).
- **DTS component** (hopping off): PSD tested against 8 dBm/3 kHz limit. Since average power was chosen for DTS, the PSD uses the average method (AVGPSD-1).

3.3 Cascading Effects of Average Power Choice

Per §15.247(d):

"...If the transmitter complies with the conducted power limits based on the use of RMS averaging over a time interval, as permitted under paragraph (b)(3) of this section, the attenuation required under this paragraph shall be 30 dB instead of 20 dB."

Per §15.247(e):

"...The same method of determining the conducted output power shall be used to determine the power spectral density."

Summary of cascading effects: - Average power → Average PSD method (AVGPSD-1 per C63.10 §11.10.3) - Average power → 30 dB band-edge attenuation (instead of 20 dB) - FHSS peak power → 20 dB band-edge attenuation

Section 2: Three-Layer Amplitude Processing Reference

A spectrum analyzer in swept mode has three independent settings that control how signal amplitude is processed. Getting the correct combination for each test is critical. The layers are:

Layer 1 — Detector Type: How raw samples within each trace point's time bucket are reduced to a single value (e.g., Positive Peak, Average/RMS, Sample).

Layer 2 — Trace Type: How data from successive sweeps is combined (e.g., Clear Write, Max Hold, Average).

Layer 3 — Avg Mode: The mathematical method used when Trace Type is set to Average (Log dB averaging, RMS power averaging, or Scalar voltage averaging).

 **CRITICAL — Log Averaging Prohibition:** C63.10 §11.12.2.5.1 explicitly states: *"Log or dB averaging shall not be used"* for average power measurements. Log Avg Mode performs exactly this prohibited operation. For power and PSD tests (1.3, 1.4, 3.1), Avg Mode **must** be set to **RMS**.

For average power/PSD tests (1.3, 1.4, 3.1), ALL THREE layers must be set: 1. Detector Type = **Average (RMS)** 2. Trace Type = **Average** 3. Avg Mode = **RMS**

Default after Preset: Detector = Normal (Auto On), Trace Type = Clear Write, Avg Mode = Log. All three must be manually changed for power tests.

Section 3: DTS Conducted Tests

Test 1.1 — DTS Bandwidth (6 dB)

Regulatory Basis

47 CFR §15.247(a)(2): "Systems using digital modulation techniques may operate in the 902-928 MHz, 2400-2483.5 MHz, and 5725-5850 MHz bands. The minimum 6 dB bandwidth shall be at least 500 kHz."

Limit: 6 dB bandwidth $\geq$ 500 kHz

C63.10-2020 Procedure — §11.8.1 (Option 1)

Verbatim from ANSI C63.10-2020, §11.8, §11.8.1:

"One of the following procedures may be used to determine the modulated DTS bandwidth. The steps for the first option are as follows:

- a) Set RBW = shall be in the range of 1% to 5% of the OBW but not less than 100 kHz.
- b) Set the VBW $\geq$ $[3 \times \text{RBW}]$.
- c) Detector = peak.
- d) Trace mode = max-hold.
- e) Sweep = No faster than coupled (auto) time.
- f) Allow the trace to stabilize.
- g) Measure the maximum width of the emission by placing two markers, one at the lowest frequency and the other at the highest frequency of the envelope of the spectral display, such that each marker is at or slightly below the '-6 dB down amplitude'. If a marker is below this '-6 dB down amplitude' value, then it shall be as close as possible to this value."

C63.10-2020 Procedure — §11.8.2 (Option 2)

Verbatim from ANSI C63.10-2020, §11.8.2:

"The automatic bandwidth measurement capability of an instrument may be employed using the X dB bandwidth mode with X set to 6 dB, if the functionality described in 11.8.1 (i.e., RBW = 100 kHz, VBW $\geq 3 \times$ RBW, and peak detector with maximum hold) is implemented by the instrumentation function. When using this capability, care shall be taken so that the bandwidth measurement is not influenced by any intermediate power nulls in the fundamental emission that might be ≥ 6 dB."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	DTS channel frequency	—
Span	~2 MHz	—
RBW	100 kHz	§11.8.1(a) — 1%–5% of OBW, not less than 100 kHz
VBW	≥ 300 kHz	§11.8.1(b) — $\geq 3 \times \text{RBW}$
Detector	Pos Peak	§11.8.1(c)
Trace	Max Hold	§11.8.1(d)
Sweep Time	Auto	§11.8.1(e)
Avg Mode	Log	Shape measurement — Log acceptable
Measurement	N dB BW, N = –6	§11.8.2

Pass Criterion: N dB BW reading ≥ 500 kHz

Test 1.2 — Occupied Bandwidth (99% Power)

Regulatory Basis

OBW is used as the integration bandwidth reference for the maximum conducted (average) output power measurement per C63.10 §11.2(c). There is no explicit FCC OBW limit in §15.247, but OBW is required as an input to subsequent tests.

C63.10-2020, §11.2(c): "The procedures provided herein for measuring the maximum conducted (average) output power assume the use of the OBW. See 6.9.3 for guidance pertaining to the definition and measurement of OBW."

C63.10-2020 Procedure — §6.9.3

The 99% occupied bandwidth is defined as the bandwidth containing 99% of the total integrated power of the transmission. The OBW measurement uses the spectrum analyzer's built-in occupied bandwidth function with 99% power ratio. Video averaging is not permitted during OBW measurement.

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	DTS channel frequency	—
Span	~2 MHz	—
RBW	100 kHz	§6.9.3
VBW	≥ 300 kHz	§6.9.3 — $\geq 3 \times \text{RBW}$
Detector	Pos Peak	—
Trace	Clear Write or Max Hold	Video averaging (Trace Average) not permitted
Avg Mode	Log	Shape measurement — Log acceptable
OBW Power Ratio	99%	§6.9.3

Record: OBW value in Hz. Required input for Test 1.3.

Test 1.3 — Maximum Conducted (Average) Output Power

Regulatory Basis

47 CFR §15.247(b)(3): "...As an alternative to a peak power measurement, compliance with the one Watt limit can be based on a measurement of the maximum conducted output power. Maximum Conducted Output Power is defined as the total transmit power delivered to all antennas and antenna elements averaged across all symbols in the signaling alphabet when the transmitter is operating at its maximum power control level. Power must be summed across all antennas and antenna elements. The average must not include any time intervals during which the transmitter is off or is transmitting at a reduced power level."

Power Method: Average (chosen per §15.247(b)(3) alternative)

Limit: +30 dBm (1 W) maximum conducted output power

C63.10-2020 Procedure — §11.9.2.2.2 (Method AVGSA-1)

Verbatim from ANSI C63.10-2020, §11.9.2.1 and §11.9.2.2.2:

§11.9.2.1: "Some regulatory agencies permit the maximum conducted (average) output power to be measured as an alternative to the maximum peak conducted output power for determining compliance to the limit. When this option is exercised, the measured power is to be referenced to the OBW rather than to the DTS bandwidth..."

When using a spectrum analyzer or EMI receiver to perform these measurements, it shall be capable of utilizing a number of measurement points in each sweep that is greater than or equal to twice the span / RBW, to set a bin-to-bin spacing of $\leq \text{RBW} / 2$ so that narrowband signals are not lost between frequency bins. If possible, configure or modify the operation of the EUT so that it transmits continuously at its maximum power control level (see 11.6).

The intent is to test at 100% duty cycle; however, a small reduction in duty cycle (to no lower than 98%) is permitted, if required by the EUT for amplitude control purposes."

§11.9.2.2.2 Method AVGSA-1: "Method AVGSA-1 uses trace averaging with the EUT transmitting at full power throughout each sweep. The procedure for this method is as follows:

- a) Set span to >1.5 times the OBW.
- b) Set RBW = 1% to 5% of the OBW, but do not exceed 1 MHz.
- c) Set VBW $\geq [3 \times \text{RBW}]$.
- d) Number of points in sweep $\geq [2 \times \text{span} / \text{RBW}]$. (This gives bin-to-bin spacing $\leq \text{RBW} / 2$, so that narrowband signals are not lost between frequency bins.)
- e) Sweep time = auto.
- f) Detector = power averaging (rms), if available. Otherwise, use sample detector mode.
- g) If transmit duty cycle $< 98\%$, use a sweep trigger with the level set to enable triggering only on full power pulses. The transmitter shall operate at the maximum power control level for the entire duration of every sweep. If the EUT transmits continuously (i.e., with no OFF intervals) or at duty cycle $\geq 98\%$, and if each transmission is entirely at the maximum power control level, then the trigger shall be set to 'free run.'
- h) Trace average at least 100 traces in power averaging (rms) mode.
- i) Compute power by integrating the spectrum across the OBW of the signal using the instrument's band power measurement function, with band limits set equal to the OBW band-edges. If the instrument does not have a band-power function, sum the spectrum levels (in power units) at intervals equal to the RBW extending across the entire OBW of the spectrum."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	DTS channel center	—
Span	$> 1.5 \times \text{OBW}$	§11.9.2.2.2(a)
RBW	1%–5% of OBW, max 1 MHz	§11.9.2.2.2(b)

PARAMETER	VALUE	C63.10 REF
VBW	$\geq 3 \times \text{RBW}$	§11.9.2.2.2(c)
Sweep Points	$\geq 2 \times \text{span} / \text{RBW}$	§11.9.2.2.2(d)
Sweep Time	Auto	§11.9.2.2.2(e)
Detector	Average (RMS)	§11.9.2.2.2(f)
Avg Mode	RMS	§11.9.2.2.2(h) — Log prohibited by §11.12.2.5.1
Avg Count	≥ 100	§11.9.2.2.2(h)
Trace	Average	§11.9.2.2.2(h)
Trigger	Free Run (for D $\geq$ 98%)	§11.9.2.2.2(g)
Measurement	Band Power across OBW	§11.9.2.2.2(i)

⚠ CRITICAL: All three amplitude processing layers must be correct. If any one is wrong — especially Avg Mode left at the default Log — the measurement will be systematically incorrect.

Pass Criterion: Band Power reading $\leq$ +30 dBm

Test 1.4 — Power Spectral Density (Average PSD — AVGPDS-1)

Regulatory Basis

47 CFR §15.247(e): "For digitally modulated systems, the power spectral density conducted from the intentional radiator to the antenna shall not be greater than 8 dBm in any 3 kHz band during any time interval of continuous transmission. This power spectral density shall be determined in accordance with the provisions of paragraph (b) of this section. The same method of determining the conducted output power shall be used to determine the power spectral density."

Limit: $\leq$ +8 dBm in any 3 kHz band

Method: Average PSD (AVGPDS-1), required because average power was chosen in Test 1.3.

C63.10-2020 Procedure — §11.10.1 and §11.10.3 (Method AVGPDS-1)

Verbatim from ANSI C63.10-2020, §11.10.1:

"Some regulatory requirements specify a conducted PSD limit within the DTS bandwidth during any time interval of continuous transmission. Such specifications require that the same method as used to determine the conducted output power shall be used to determine the power spectral density. If

maximum peak conducted output power was measured, then the peak PSD procedure 11.10.2 (method PKPSD) shall be used. If maximum conducted output power was measured, then one of the average PSD procedures shall be used, as applicable..."

Verbatim from §11.10.3 Method AVGPSSD-1:

"Method AVGPSSD-1 uses trace averaging with EUT transmitting at full power throughout each sweep.

The following procedure may be used when the maximum (average) conducted output power was used to determine compliance to the fundamental output power limit...

- a) Set instrument center frequency to DTS channel center frequency.
- b) Set span to > 1.5 times the OBW.
- c) Set RBW to: $3 \text{ kHz} \leq \text{RBW} \leq 100 \text{ kHz}$.
- d) Set VBW $\geq [3 \times \text{RBW}]$.
- e) Detector = power averaging (rms) or sample detector (when rms not available).
- f) Ensure that the number of measurement points in the sweep $\geq [2 \times \text{span} / \text{RBW}]$.
- g) Sweep time = auto couple.
- h) Employ trace averaging (rms) mode over a minimum of 100 traces.
- i) Use the peak marker function to determine the maximum amplitude level.
- j) If the measured value exceeds requirement, then reduce RBW (but no less than 3 kHz) and repeat (note that this might require zooming in on the emission of interest and reducing the span to meet the minimum measurement point requirement as the RBW is reduced)."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	DTS channel center	§11.10.3(a)
Span	$> 1.5 \times \text{OBW}$	§11.10.3(b)
RBW	3 kHz (start conservative)	§11.10.3(c) — $3 \text{ kHz} \leq \text{RBW} \leq 100 \text{ kHz}$
VBW	$\geq 3 \times \text{RBW}$	§11.10.3(d)
Detector	Average (RMS)	§11.10.3(e) — must use if available
Sweep Points	$\geq 2 \times \text{span} / \text{RBW}$	§11.10.3(f)

PARAMETER	VALUE	C63.10 REF
Sweep Time	Auto	§11.10.3(g)
Avg Mode	RMS	§11.10.3(h)
Avg Count	≥ 100	§11.10.3(h)
Trace	Average	§11.10.3(h)
Measurement	Peak marker	§11.10.3(i)

⚠ CRITICAL: Same three-layer requirement as Test 1.3. A Log-mode reading may falsely exceed the +8 dBm limit. If the reading exceeds +8 dBm, verify all three settings before concluding non-compliance.

Pass Criterion: Peak marker reading ≤ +8 dBm (with RBW = 3 kHz, this directly reads as dBm/3 kHz)

Test 1.5 — Band-Edge Emissions (Non-Restricted Bands)

Regulatory Basis

47 CFR §15.247(d): "In any 100 kHz bandwidth outside the frequency band in which the spread spectrum or digitally modulated intentional radiator is operating, the radio frequency power that is produced by the intentional radiator shall be at least 20 dB below that in the 100 kHz bandwidth within the band that contains the highest level of the desired power...If the transmitter complies with the conducted power limits based on the use of RMS averaging over a time interval, as permitted under paragraph (b)(3) of this section, the attenuation required under this paragraph shall be 30 dB instead of 20 dB."

Limit: Since average power was chosen: emissions in any 100 kHz BW outside 902–928 MHz must be ≥ **30 dB** below the highest in-band 100 kHz PSD.

C63.10-2020 Procedure — §11.11.2 and §11.11.3

The band-edge measurement is a relative measurement comparing in-band PSD (in 100 kHz bandwidth) to out-of-band emissions (in 100 kHz bandwidth).

§11.11.2 — In-band reference level: RBW = 100 kHz, VBW ≥ 300 kHz, Peak detector, Max-hold trace, span ≥ 1.5× DTS bandwidth. Peak marker finds the highest in-band 100 kHz level.

§11.11.3 — Band-edge emission measurement: Same RBW/VBW/detector/trace settings. Span widened to capture both the in-band peak and emissions outside the authorized band. Marker-delta function used to measure the dB difference between in-band peak and highest out-of-band emission.

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	Lowest DTS channel (near 902 MHz edge)	§11.11 — test at lowest and highest channels
Span (phase 1)	$\geq 1.5 \times \text{DTS BW}$	§11.11.2
RBW	100 kHz	§11.11.2
VBW	$\geq 300 \text{ kHz}$	§11.11.2
Detector	Pos Peak	§11.11.2
Trace	Max Hold	§11.11.2
Avg Mode	Log	Relative measurement — Log acceptable
Span (phase 2)	Widen to show band edge	§11.11.3
Measurement	Delta marker: in-band peak to highest out-of-band emission	§11.11.3

Pass Criterion: Delta marker $\leq -30 \text{ dB}$. Repeat for highest DTS channel (near 928 MHz).

Section 4: FHSS Conducted Tests

Test 2.1 — 20 dB Bandwidth

Regulatory Basis

47 CFR §15.247(a)(1): "Frequency hopping systems shall have hopping channel carrier frequencies separated by a minimum of 25 kHz or the 20 dB bandwidth of the hopping channel, whichever is greater."

47 CFR §15.247(a)(1)(i): "...if the 20 dB bandwidth of the hopping channel is less than 250 kHz, the system shall use at least 50 hopping frequencies...if the 20 dB bandwidth of the hopping channel is 250 kHz or greater, the system shall use at least 25 hopping frequencies... The maximum allowed 20 dB bandwidth of the hopping channel is 500 kHz."

Limit: 20 dB BW $\leq$ 500 kHz. Also determines minimum hopping channel count (50 if < 250 kHz, 25 if ≥ 250 kHz).

C63.10-2020 Procedure

The 20 dB bandwidth measurement follows the same general methodology as the DTS 6 dB bandwidth (§11.8) but with $N = 20$ dB. Frequency hopping shall be disabled for this test.

Instrument Settings Summary

PARAMETER	VALUE	NOTES
Center Freq	FHSS channel frequency	Hopping disabled, single channel
Span	~1 MHz	~5× expected OBW for 125 kHz LoRa
RBW	5 kHz	~1% of OBW for fine resolution
VBW	≥ 20 kHz	$\geq 3 \times$ RBW
Detector	Pos Peak	—
Trace	Max Hold	—
Avg Mode	Log	Shape measurement
Measurement	N dB BW, $N = -20$	—

Pass Criterion: 20 dB BW $\leq$ 500 kHz. Record value — needed for Test 2.2 and to determine required hopping channel count.

Test 2.2 — Carrier Frequency Separation

Regulatory Basis

47 CFR §15.247(a)(1): "Frequency hopping systems shall have hopping channel carrier frequencies separated by a minimum of 25 kHz or the 20 dB bandwidth of the hopping channel, whichever is greater."

Limit: Channel separation $\geq \max(25 \text{ kHz}, 20 \text{ dB BW from Test 2.1})$

C63.10-2020 Procedure — §7.8.2

Verbatim from ANSI C63.10-2020, §7.8.2:

"The EUT shall have its hopping function enabled. Use the following spectrum analyzer settings:

- a) Span: Wide enough to capture the peaks of two adjacent channels.
- b) RBW: Start with the RBW set to approximately 30% of the channel spacing; adjust as necessary to best identify the center of each individual channel.
- c) Video (or average) bandwidth (VBW) $\geq$ RBW.
- d) Sweep: No faster than coupled (auto) time.
- e) Detector function: Peak.
- f) Trace: Max-hold.
- g) Allow the trace to stabilize.

Use the marker-delta function to determine the separation between the peaks of the adjacent channels. Compliance of an EUT with the appropriate regulatory limit shall be determined. A spectral plot of the data shall be included in the test report."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	Midpoint between two adjacent channels	§7.8.2(a)
Span	~500 kHz	§7.8.2(a) — capture both channel peaks
RBW	~30% of channel spacing	§7.8.2(b)
VBW	$\geq$ RBW	§7.8.2(c)
Sweep Time	Auto	§7.8.2(d)
Detector	Pos Peak	§7.8.2(e)

PARAMETER	VALUE	C63.10 REF
Trace	Max Hold	§7.8.2(f)
Avg Mode	Log	Shape measurement
Measurement	Delta marker: frequency separation between adjacent channel peaks	§7.8.2

Pass Criterion: Frequency separation $\geq \max(25 \text{ kHz}, 20 \text{ dB BW from Test 2.1})$

Test 2.3 — Peak Conducted Output Power (FHSS)

Regulatory Basis

47 CFR §15.247(b)(2): "For frequency hopping systems operating in the 902-928 MHz band: 1 watt for systems employing at least 50 hopping channels; and, 0.25 watts for systems employing less than 50 hopping channels, but at least 25 hopping channels..."

Power Method: Peak only. The average alternative in §15.247(b)(3) applies only to "systems using digital modulation" — paragraph (b)(2) does not offer this alternative for FHSS.

Limit: +30 dBm (1 W) peak for ≥ 50 hopping channels; +24 dBm (0.25 W) peak for 25–49 channels.

C63.10-2020 Procedure — §7.8.5

Verbatim from ANSI C63.10-2020, §7.8.5:

"This is an RF-conducted test to evaluate maximum peak output power. Use a direct connection between the antenna port of the unlicensed wireless device and the spectrum analyzer, through suitable attenuation. Frequency hopping shall be disabled for this test. Use the following spectrum analyzer settings:

- a) Span: Approximately five times the 20 dB bandwidth, centered on a hopping channel.
- b) RBW > 20 dB bandwidth of the emission being measured.
- c) VBW $\geq$ RBW.
- d) Sweep: No faster than coupled (auto) time.
- e) Detector function: Peak.
- f) Trace: Max-hold.

- g) Allow trace to stabilize.
- h) Use the marker-to-peak function to set the marker to the peak of the emission.
- i) The indicated level is the peak output power, after any corrections for external attenuators and cables.
- j) A spectral plot of the test results and setup description shall be included in the test report."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	FHSS channel frequency	— Hopping disabled
Span	$\sim 5 \times 20$ dB BW	§7.8.5(a)
RBW	> 20 dB BW	§7.8.5(b)
VBW	$\geq$ RBW	§7.8.5(c)
Sweep Time	Auto	§7.8.5(d)
Detector	Pos Peak	§7.8.5(e)
Trace	Max Hold	§7.8.5(f)
Measurement	Peak marker	§7.8.5(h)(i)

Pass Criterion: Marker reading $\leq +30$ dBm (for ≥ 50 hopping channels)

Test 2.4 — Occupied Bandwidth (FHSS)

Regulatory Basis

OBW is used as a reference for other FHSS measurements. No explicit FCC limit on FHSS OBW.

C63.10-2020 Procedure — §7.8.6

Verbatim from ANSI C63.10-2020, §7.8.6:

"For occupied bandwidth measurements, use the procedure in 6.9.3. Frequency hopping shall be disabled for this test."

Apply the same OBW procedure as Test 1.2 to the FHSS channel with hopping disabled.

Test 2.5 — Band-Edge Emissions (Static, Hopping Off)

Regulatory Basis

Same as Test 1.5 — §15.247(d). For FHSS under peak power, the attenuation requirement is **20 dB**.

C63.10-2020 Procedure — §7.8.7.1 and §7.8.7.2

Verbatim from ANSI C63.10-2020, §7.8.7.1:

"To demonstrate compliance with the relative out-of-band emissions requirements conducted spurious emissions shall be measured for the transmit frequencies, per 5.5 and 5.6, and at the maximum transmit powers. Frequency hopping shall be disabled for this test with the exception of measurements at the allocated band-edges which shall be repeated with hopping enabled.

Connect the primary antenna port through an attenuator to the spectrum analyzer input...The frequency range of testing shall span 30 MHz to 10 times the operating frequency...The resolution bandwidth shall be 100 kHz, video bandwidth 300 kHz, and a coupled sweep time with a peak detector.

The limit is based on the highest in-band level across all channels measured using the same instrument settings (resolution bandwidth of 100 kHz, video bandwidth of 300 kHz, and a coupled sweep time with a peak detector)."

Verbatim from §7.8.7.2:

"Compliance with a relative limit at the band-edges (e.g., -20 dBc) shall be made on the lowest and on the highest channels with frequency hopping disabled and repeated with frequency hopping enabled. For the latter test the hopping sequence shall include the lowest and highest channels.

For measurements with the hopping disabled the analyzer screen shall clearly show compliance with the requirement within 10 MHz of the allocated band-edge."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	Lowest FHSS channel	§7.8.7.2 — Hopping disabled
Span	Wide enough to show channel + emissions within 10 MHz of band edge	§7.8.7.2
RBW	100 kHz	§7.8.7.1
VBW	300 kHz	§7.8.7.1
Sweep Time	Auto (coupled)	§7.8.7.1

PARAMETER	VALUE	C63.10 REF
Detector	Pos Peak	§7.8.7.1
Trace	Max Hold	—
Avg Mode	Log	Relative measurement
Measurement	Delta marker: in-band peak to highest out-of-band emission	§7.8.7.1

Pass Criterion: Delta $\leq$ -20 dB (FHSS peak power $\rightarrow$ 20 dB requirement). Repeat for highest FHSS channel.

Test 2.6 — Band-Edge Emissions (Hopping On)

C63.10-2020 Procedure — §7.8.7.2 (continued)

Same setup as Test 2.5, but with hopping enabled. Per §7.8.7.2: "repeated with frequency hopping enabled. For the latter test the hopping sequence shall include the lowest and highest channels."

Allow Max Hold trace sufficient time to capture all hop channels including edge channels before reading results.

Pass Criterion: Same as Test 2.5 — delta $\leq$ -20 dB

Test 2.7 — Conducted Spurious Emissions

C63.10-2020 Procedure — §7.8.7.1

Per §7.8.7.1: "The frequency range of testing shall span 30 MHz to 10 times the operating frequency." For 915 MHz, this means 30 MHz to ~9.15 GHz.

Settings per §7.8.7.1: RBW = 100 kHz, VBW = 300 kHz, coupled sweep time, peak detector.

First establish the in-band reference level (highest in-band emission in 100 kHz BW). Then sweep the full frequency range in sub-sweeps as needed to cover 30 MHz to the instrument's maximum frequency. Compare all spurious emissions to the in-band reference.

⚠ **NOTE:** C63.10 requires testing to 10× operating frequency (~9.15 GHz for 915 MHz operation). Verify your instrument's maximum frequency range is sufficient. If not, document this limitation and plan for supplemental testing with a wider-range instrument for formal certification.

Pass Criterion: All spurious emissions ≥ 20 dB below in-band reference level (FHSS peak power path)

Test 2.8 — Number of Hopping Frequencies

Regulatory Basis

47 CFR §15.247(a)(1)(i): "...the system shall use at least 50 hopping frequencies..." (for 20 dB BW < 250 kHz)

Limit: ≥ 50 hopping channels (for 20 dB BW < 250 kHz); ≥ 25 (for 20 dB BW ≥ 250 kHz)

C63.10-2020 Procedure — §7.8.3

Verbatim from ANSI C63.10-2020, §7.8.3:

"The EUT shall have its hopping function enabled. Use the following spectrum analyzer settings:

- a) Span: The frequency band of operation. Depending on the number of channels the device supports, it could be necessary to divide the frequency range of operation across multiple spans, to allow the individual channels to be clearly seen.
- b) RBW: To identify clearly the individual channels, set the RBW to less than 30% of the channel spacing or the 20 dB bandwidth, whichever is smaller.
- c) VBW $\geq$ RBW.
- d) Sweep: No faster than coupled (auto) time.
- e) Detector function: Peak.
- f) Trace: Max-hold.
- g) Allow the trace to stabilize.

It might prove necessary to break the span up into subranges to show clearly all of the hopping frequencies. Compliance of an EUT with the appropriate regulatory limit shall be determined for the number of hopping channels. A spectral plot of the data shall be included in the test report."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Center Freq	Center of FHSS band	§7.8.3(a)
Span	Full FHSS band of operation	§7.8.3(a)
RBW	< 30% of channel spacing or 20 dB BW (whichever smaller)	§7.8.3(b)
VBW	≥ RBW	§7.8.3(c)
Sweep Time	Auto	§7.8.3(d)
Detector	Pos Peak	§7.8.3(e)
Trace	Max Hold	§7.8.3(f)
Avg Mode	Log	Shape measurement
Measurement	Count distinct peaks (use peak table if available)	§7.8.3

Pass Criterion: Peak count ≥ 50 (for 20 dB BW < 250 kHz)

Test 2.9 — Time of Occupancy (Dwell Time)

Regulatory Basis

47 CFR §15.247(a)(1)(i): "...the average time of occupancy on any frequency shall not be greater than 0.4 seconds within a 20 second period" (for 20 dB BW < 250 kHz with ≥ 50 channels)

Limit: Average dwell ≤ 0.4 seconds per channel per 20-second observation window

C63.10-2020 Procedure — §7.8.4

Verbatim from ANSI C63.10-2020, §7.8.4:

"The dwell time per hop on a channel is the time from the start of the first transmission to the end of the last transmission for that hop. If the device has a single transmission per hop then the dwell time is the duration of that transmission.

Use the following spectrum analyzer settings to determine the dwell time per hop:

- Zero span mode centered on a single hopping channel.
- RBW: Greater than the 20 dB bandwidth. Where this is not possible, a smaller RBW may be used, provided that the transient response of the analyzer has settled during the time of the transmission.

c) Setting the sweep time to be slightly longer than the hopping period per channel (hopping period = 1/hopping rate) should achieve this.

d) Use a video trigger, where possible with a trigger delay, so that the start of the transmission is clearly observed. The trigger level might need adjustment to reduce the chance of triggering when the system hops on an adjacent channel.

e) Detector function: Peak.

f) Trace: Clear-write, single sweep.

g) Place markers at the start of the first transmission on the channel and at the end of the last transmission. The dwell time per hop is the time between these two markers."

Instrument Settings Summary

PARAMETER	VALUE	C63.10 REF
Mode	Zero span	§7.8.4(a)
Center Freq	Single FHSS channel	§7.8.4(a)
RBW	> 20 dB BW	§7.8.4(b)
VBW	≥ RBW	—
Sweep Time	Slightly longer than hop period (manual)	§7.8.4(c)
Trigger	Video trigger	§7.8.4(d)
Detector	Pos Peak	§7.8.4(e)
Trace	Clear Write, single sweep	§7.8.4(f)
Measurement	Delta marker: start to end of burst	§7.8.4(g)

Pass Criterion: Dwell time ≤ 0.4 seconds per hop

Section 5: Hybrid Conducted Tests

Regulatory Basis for All Hybrid Tests

47 CFR §15.247(f): "For the purposes of this section, hybrid systems are those that employ a combination of both frequency hopping and digital modulation techniques. The frequency hopping operation of the hybrid system, with the direct sequence or digital modulation operation turned-off, shall have an average time of occupancy on any frequency not to exceed 0.4 seconds within a time period in seconds equal to the number of hopping frequencies employed multiplied by 0.4. The power spectral density conducted from the intentional radiator to the antenna due to the digital modulation operation of the hybrid system, with the frequency hopping operation turned off, shall not be greater than 8 dBm in any 3 kHz band during any time interval of continuous transmission."

Hybrid testing decomposes the system into its two components: - **DTS component** (hopping off):
PSD $\leq$ 8 dBm/3 kHz - **FHSS component** (DTS off): Dwell time and channel count requirements

Test 3.1 — Power Spectral Density (Hybrid, AVGPSD-1)

Regulatory Basis

Per §15.247(f): PSD of the digital modulation component (hopping off) $\leq$ 8 dBm in any 3 kHz band.
Per §15.247(e): the same power measurement method must be used for PSD — since average was chosen for DTS, use AVGPSD-1.

C63.10-2020 Procedure — §11.10.3 (AVGPSD-1)

Same procedure as Test 1.4. Refer to Test 1.4 for the complete C63.10 text.

DUT configuration: Frequency hopping OFF, DTS modulation ON (per §15.247(f))

Pass Criterion: Peak marker reading $\leq$ +8 dBm (with RBW = 3 kHz)

Test 3.2 — Band-Edge Emissions (Hybrid Static, Hopping Off)

Regulatory Basis

Same band-edge requirement as DTS tests — §15.247(d). Since average power is used, attenuation is **30 dB**.

DUT configuration: Hopping OFF, DTS modulation ON.

Same procedure as Test 1.5.

Pass Criterion: Delta $\leq$ -30 dB

Test 3.3 — Band-Edge Emissions (Hybrid, Hopping On)

Regulatory Basis

Same as Test 3.2 but with FHSS component active. Both hopping and DTS modulation are active.

Same setup as Test 3.2 but with hopping enabled. Allow Max Hold trace sufficient time to capture all sub-band hop channels before reading results.

Pass Criterion: Delta $\leq$ -30 dB (average power path $\rightarrow$ 30 dB requirement)

Test 3.4 — Number of Hopping Channels (Hybrid)

Regulatory Basis

Per §15.247(f), the FHSS component of the hybrid must meet the hopping requirements of §15.247(a)(1)(i).

DUT configuration: DTS OFF, FHSS ON (per §15.247(f))

Same procedure as Test 2.8, adjusted for the hybrid sub-band channel count and frequency range.

Pass Criterion: Channel count meets minimum per §15.247(a)(1)(i)

Test 3.5 — Time of Occupancy (Hybrid)

Regulatory Basis

Per §15.247(f): "average time of occupancy on any frequency not to exceed 0.4 seconds within a time period in seconds equal to the number of hopping frequencies employed multiplied by 0.4."

For 8 channels: observation window = $8 \times 0.4 = 3.2$ seconds.

DUT configuration: DTS OFF, FHSS ON (per §15.247(f))

Same procedure as Test 2.9. Set sweep time to slightly longer than the observation window (e.g., 3.5 s for 8 channels).

Pass Criterion: Dwell $\leq$ 0.4 seconds per channel within the observation window

Section 6: Quick-Reference Settings Matrix

TEST	DESCRIPTION	DETECTOR	TRACE TYPE	AVG MODE	AVG COUNT
1.1	DTS 6 dB BW	Pos Peak	Max Hold	Log	N/A
1.2	DTS OBW 99%	Pos Peak	Clear Write or Max Hold	Log	N/A
1.3	DTS Average Power	Average (RMS)	Average	RMS	≥ 100
1.4	DTS Average PSD (AVGPSD-1)	Average (RMS)	Average	RMS	≥ 100
1.5	DTS Band-Edge	Pos Peak	Max Hold	Log	N/A
2.1	FHSS 20 dB BW	Pos Peak	Max Hold	Log	N/A
2.2	FHSS Carrier Separation	Pos Peak	Max Hold	Log	N/A
2.3	FHSS Peak Power	Pos Peak	Max Hold	RMS (best practice)	N/A
2.4	FHSS OBW	Pos Peak	Clear Write or Max Hold	Log	N/A
2.5	FHSS Band-Edge (static)	Pos Peak	Max Hold	Log	N/A
2.6	FHSS Band-Edge (hopping)	Pos Peak	Max Hold	Log	N/A
2.7	FHSS Spurious	Pos Peak	Max Hold	Log	N/A
2.8	FHSS Hop Count	Pos Peak	Max Hold	Log	N/A
2.9	FHSS Dwell Time	Pos Peak	Clear Write	Log	N/A
3.1	Hybrid PSD (AVGPSD-1)	Average (RMS)	Average	RMS	≥ 100
3.2	Hybrid Band-Edge (static)	Pos Peak	Max Hold	Log	N/A
3.3	Hybrid Band-Edge (hopping)	Pos Peak	Max Hold	Log	N/A
3.4	Hybrid Hop Count	Pos Peak	Max Hold	Log	N/A
3.5	Hybrid Dwell Time	Pos Peak	Clear Write	Log	N/A

Bold entries indicate settings that differ from typical analyzer defaults and must be explicitly set.

Document generated from cross-reference of 47 CFR §15.247 and ANSI C63.10-2020 with Corrigendum 1 (2023).

C63.10-2020 Corrigendum 1 scope verified: addresses LISN validation removal, ANSI C63.30 references, 10 dB emission BW formula correction, and Equation L.1 correction. No changes affect DTS, FHSS, or hybrid test procedures used in this document.